Thermal effects and energy budgets

antocciphysics | TEACHER KEY | CPA

Learning target: Estimate a temperature change only after stating how much energy enters the material.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 8; 3 course days later.*
Work input enables the transfer; it does not contradict the direction of spontaneous heat flow.

02. Explain a worked example (worked)
Absorbed 250 J heats the surface under the stated model; reflected/transmitted portions go elsewhere.

03. Look and explain (model)
20 J absorbed; 80 J elsewhere.

04. Build the solution (practice)
2°C.

05. Use the idea (reason)
Phase-change latent heat, changed properties, energy transfer/losses; constant-c one-phase model no longer adequate.

06. Choose a method (mixed) *Review: Unit 4, Day 21; 12 course days later.*
18 J other; fraction 0.70 or 70%.

07. Exit check (exit)
Doubles.

08. STRETCH: Apply it another way (stretch)
Larger mass has half the rise for the same absorbed heat and c, absent phase change/losses.

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Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 8; 3 course days later.*
Total energy is conserved. At common temperature the initial temperature-difference opportunity for work is eliminated.

02. Explain a worked example (worked)
Absorbed 250 J heats the surface under the stated model; reflected/transmitted portions go elsewhere.

03. Represent and interpret (model)
20 J absorbed; 80 J elsewhere.

04. Solve and justify (practice)
Absorbed 20, 000 J;ΔT 50°C.

05. Analyze the reasoning (reason)
Phase-change latent heat, changed properties, energy transfer/losses; constant-c one-phase model no longer adequate.

06. Choose a method (mixed) *Review: Unit 4, Day 21; 12 course days later.*
Initial U=240 J; K=232 J; v≈10.77 m/s.

07. Exit check (exit)
8000 J; 200, 000 J.

08. STRETCH: Apply it another way (stretch)
ΔU=900 J; ΔT=900/180=5 K.